Introduction to Large Language Models
Assignment - 7

Number of Questions: 8
Total Marks: 6 × 1 + 2 × 2 = 10

QUESTION 1: [1 mark]

Which of the following statements correctly differentiates the pre-training paradigms of **BERT** and **T5**?

A. BERT uses masked language modeling with bidirectional attention, whereas T5 utilizes a span corruption objective within an encoder-decoder architecture.
B. BERT uses next-token prediction, whereas T5 uses next-sentence prediction.
C. BERT is an encoder-decoder model, whereas T5 is a decoder-only model.
D. BERT uses span corruption on C4, whereas T5 is trained on BookCorpus using a causal decoder.

Correct Answer: A
Solution: BERT is an encoder-only architecture trained with Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). In contrast, T5 follows a sequence-to-sequence encoder-decoder setup trained to reconstruct corrupted contiguous spans using sentinel tokens.

QUESTION 2: [1 mark]

Which attention mechanism and masking scheme are natively employed during the auto-regressive generation phase in a Transformer **decoder** to prevent attending to future token positions?

A. Bidirectional attention with a fully-visible mask
B. Causal (autoregressive) self-attention with a lower-triangular mask
C. Prefix-LM mask allowing unrestricted visibility across all generated tokens
D. Cross-attention with a diagonal identity mask

Correct Answer: B
Solution: Auto-regressive decoders utilize causal self-attention enforced via a lower-triangular mask (causal mask), which masks future positions with negative infinity so that tokens can only attend to previous and current positions.

QUESTION 3: [1 mark]

Which of the following pre-training corpora was curated specifically for pre-training the **T5** model to provide a cleaner, highly-filtered alternative to raw web scrapes?

A. BookCorpus

B. Wikipedia-100M
C. Colossal Clean Crawled Corpus (C4)
D. OpenWebText

Correct Answer: C

Solution: The C4 (Colossal Clean Crawled Corpus) dataset was introduced by Raffel et al. specifically for T5 to mitigate noisy, boilerplate, and low-quality text present in raw Common Crawl dumps.

QUESTION 4: [1 mark]

In the ELMo task-specific combination formula, what is the primary function of the softmax-normalized scalar weights s_j^{task}?

A. They rescale the output embedding dimensionality to match downstream architectures.
B. They determine the relative task-specific importance and contribution of each internal bi-LSTM layer representation.
C. They prevent gradient explosion during the character-CNN convolution step.
D. They linearly combine the forward and backward language model losses during unsupervised pre-training.

Correct Answer: B

Solution: In ELMo's downstream adaptation formula $ELMo_k^{task} = \gamma^{task} \Sigma_j s_j^{task} h_{k,j}^{LM}$, the parameters s_j^{task} are softmax-normalized weights that enable downstream tasks to learn the relative contribution of each representation layer (e.g., lower syntactic layers versus higher semantic layers).

QUESTION 5: [1 mark]

Which model architecture family is natively suited for sequence-to-sequence tasks such as **Machine Translation** and **Abstractive Summarization** without requiring structural modifications or prompt-based workarounds?

A. Encoder-only models (e.g., BERT, RoBERTa)
B. Encoder-Decoder models (e.g., BART, T5)
C. Character-level CNNs with unidirectional LSTMs
D. Pure classifier heads with static word embeddings

Correct Answer: B

Solution: Sequence-to-sequence encoder-decoder architectures naturally map arbitrary input sequences *x* to target output sequences *y* (modeling *P(y|x)*) through a bidirectional encoder and an autoregressive decoder.

QUESTION 6: [1 mark]

What major modifications were introduced in the **GPT-2** architecture and training recipe

compared to the original **GPT-1**?

A. Relocating layer normalization to the input of each sub-block (pre-LN) and adding a layer norm after the final self-attention block.
B. Expanding the vocabulary size to ~50,257 and increasing the context window to 1024 tokens.
C. Scaling residual layer weights at initialization by a factor of $1/\sqrt{(2N)}$.
D. All of the above.

Correct Answer: D

Solution: GPT-2 introduced pre-layer normalization, added a final layer norm, modified residual initialization scaling, expanded context length to 1024, and broadened its BPE vocabulary to 50,257 tokens.

QUESTION 7: [2 marks]

Which of the following statement(s) regarding token representations and pre-training objectives is/are **TRUE**?

A. ELMo employs a character-level CNN at the input layer to handle out-of-vocabulary (OOV) tokens and capture morphological subword patterns.
B. BERT generates dynamic, context-dependent token representations using Masked Language Modeling and Next Sentence Prediction.
C. Standard autoregressive decoder-only models maximize the bidirectional context likelihood $\prod_{t=1}^{T} P(w_t \mid w_1, ..., w_T)$.
D. BART incorporates a bidirectional encoder with an autoregressive decoder and trains on various noising schemes including token deletion and sentence permutation.

Correct Answer: A, B, D

Solution:

• *(A) is True:* ELMo's character-level CNN constructs word vectors from character sequences, eliminating out-of-vocabulary tokens.
• *(B) is True:* BERT computes bidirectional contextual embeddings via masked language modeling and sentence pair conditioning.
• *(C) is False:* Decoder-only models are autoregressive and factorize probabilities causally: $\prod_{t=1}^{T} P(w_t \mid w_{1:t-1})$.
• *(D) is True:* BART uses multiple corruption strategies including sentence shuffling, token masking, span deletion, and text infilling.

QUESTION 8: (Numerical Question) [2 marks]

The einsum function in NumPy is used as a generalized operation for tensor contractions. Consider the two matrices:

Matrix A	Matrix B
&bmatrix; 4 -2 3 1 &bmatrix;	&bmatrix; 5 2 -3 6 &bmatrix;

What is the scalar output of the following NumPy operation?

```
numpy.einsum('ij,ji->', A, B)
```

Correct Answer: 38

Solution: The Einstein summation subscript 'ij,ji->' represents the contraction $\Sigma_i \Sigma_j A_{ij} B_{ji}$, which evaluates to:

$(A_{00} \times B_{00}) + (A_{01} \times B_{10}) + (A_{10} \times B_{01}) + (A_{11} \times B_{11})$

$= (4 \times 5) + (-2 \times -3) + (3 \times 2) + (1 \times 6)$

$= 20 + 6 + 6 + 6 = 38$